

Freight Rate Prediction Challenge

Machine Learning Assessment Report

Objective. Train a model on the labeled development data, validate the modeling approach without using the final validation labels, generate predictions for all 12,000 validation loads, and produce the required December 2025 prediction chart.

1. Data exploration and quality

The development dataset contains 48,000 labeled loads covering January through October 2025. The final validation dataset contains 12,000 loads from November through December 2025. The development data contains missing values in weight (300 rows) and market_index (374 rows). The final model uses fields that are also available in the fixed December input file, avoiding a feature-availability mismatch.

2. Validation and split

A chronological holdout was used: January-August 2025 for model fitting and September-October 2025 for validation. This mirrors the actual forecasting setting because the unlabeled validation period occurs after the labeled development period. Random splitting was avoided because it could mix future-like observations into training and make the estimated generalization error overly optimistic.

3. Feature engineering

The model uses pickup, delivery, distance, equipment and weight, plus a route identifier and weight-per-mile. Calendar features include year, month, day of month, day of week, day of year and ISO week. The load_id is excluded because it is an identifier rather than a predictive business feature. CatBoost handles numeric missing values natively; categorical missing values are mapped to a dedicated MISSING category.

4. Model selection

CatBoost regression was selected after comparing tree-based alternatives on the chronological holdout. The final model uses depth 7, learning rate 0.04, L2 regularization 8 and early stopping during development. The final fit is retrained on all 48,000 labeled rows using the selected iteration count.

5. Holdout performance

Metric	September-October holdout
RMSE	633.69
MAE	122.17
R ²	0.8276
Selected iterations	178

6. Final predictions

The final model is trained on all labeled development data and applied to every row in validation.csv. The template load_id order is preserved, and the resulting file contains exactly load_id,predicted_rate. The same trained model is applied to the 31 fixed December rows, where only the date changes.

7. Reproducibility and submission

The repository includes the training script, scorer, requirements, README, model artifact and generated outputs. The supplied scorer validates the 12,000 validation predictions and 31 December predictions and creates candidate_december.png.

December 2025 prediction chart

